

Holt 2.0

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1 Introduction

This package implements the forecasting method devised by C. C. Holt (?), namely double exponential smoothing with level and trend and components. It requires time-series data. If you want triple exponential smoothing (with a seasonal component), as in (?), try the Winters package.

A prior version of this package (1.0) was written by Ignacio Díaz-Emparanza in 2015. This revision was written by Allin Cottrell, exploiting modern `hansl` syntax and adding functionality for automatic determination of the Holt smoothing parameters. See Section ?? below for help with updating scripts for use with version 2.

The filtering recursions and forecasting equation for Holt’s method are shown below. The notation is based on ?: ℓ denotes the level and b the trend or slope; α and β are the smoothing parameters.

$$\begin{aligned}\ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1} \\ \hat{y}_{t+h|t} &= \ell_t + hb_t\end{aligned}$$

Like most functions in `gretl`, `Holt` confines its operations to the sample range in force when the function is called. We’ll refer to the (1-based) start and end of this range as t_1 and t_2 , respectively.¹ In addition, let T denote the date of the last valid observation of the variable to be predicted, y .

There are two cases to be considered:

1. $T < t_2$: in this case there’s a ‘tail’ of observations for which values of y_t are not available. Winters will by default produce predictions for these observations.
2. $T = t_2$: there’s no tail for genuine out-of-sample prediction, but you can specify a number of observations, `ntest`, to reserve for pseudo-out-of-sample prediction. In that case the Holt training recursion will be terminated at observation $T - \text{ntest}$.

If you want case 1 but your dataset doesn’t contain any extra trailing observations it’s straightforward to add them using the `dataset addobs` command, or via the Add observations item under the Data menu in the graphical interface.

Conversely, if the dataset does contain extra trailing observations but you want case 2, specifying a positive value for `ntest` will result in the extra observations being ignored and pseudo-out-of-sample forecasts being produced.

2 The public functions

The primary function has the following signature:

```
function bundle Holt (const series y, const bundle specs[null])
```

¹This range can be set via the `smp1` command. If `smp1` has not been invoked since a dataset was opened t_1 will equal 1 and t_2 will equal the total number of observations.

The required argument `y` specifies the series to be processed. The second argument—which can be omitted, as indicated by the `null` modifier—may contain one or more of the following options:

- `theta` (matrix): if supplied, this must be a 2-vector holding the smoothing parameters $\theta = (\alpha, \beta)$, with each parameter in $[0, 1]$. If `parms` is not given these parameters are determined automatically.
- `ntest` (integer): specifies the number of observations at the end of the range of `y` values to reserve for testing via pseudo-out-of-sample forecasting. The default value is 0.
- `verbose` (integer): controls the verbosity of output, with levels 0, 1 (the default), 2 or 3.

The return value from `Holt` is a bundle containing the following elements:

- `theta`: a 2-vector holding the smoothing parameter values.
- `init`: a 2-vector of holding the initializer: level and trend.
- `ntest`: a non-negative integer indicating how many observations were reserved for pseudo-out-of-sample forecasting.
- `W`: a $T \times 4$ matrix. Column 1 holds the data passed as `y`; columns 2 and 3 hold the estimated level and trend; and column 4 holds forecasts. More details on this matrix are provided below.
- `RMSE_train`: the root mean squared error for the 1-step ahead forecasts made using the training data.

If `ntest` is non-zero, two additional items will be included in the bundle, namely the RMSE and Mean Absolute Error for the pseudo-out-of-sample forecasts, under the keys `RMSE_test` and `MAE_test`.

A simple plotting function is also supplied:

```
function void Holt_plot (bundle *b, int ptype)
```

The first argument should be a bundle obtained from `Holt` and the second argument should be either 1 for a plot of actual and forecast values, or 2 for a three-part plot of the level, trend and irregular components.

3 Command-line usage

We'll start with a simple example [FIXME FIND ONE!] replication of the Australian tourism example given by ?. Here's the `hansl` code:

```
open XXX.gdt
dataset addobs 12
bundle hb = Holt(XXX)
```

As an alternative to the above one might not add any observations but use the `ntest` option to get Winters to generate pseudo-out-of-sample forecasts:

```
open XXX.gdt #- -frompkg=Winters
bundle wb = Holt(y, _(ntest=8))
```

This will produce forecasts for 2016:1 to 2017:4.